

A Field Project Report
on
**DATA VISUALIZATION OF PUBLIC HEALTH &
ENVIRONMENTAL DATA**

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2025 – 26

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DECLARATION

We 9040 - Purva Kante, 9039 – Sachi Kangutkar, 9022 – Shravani Ghode, 9039 – Akshata Khambe, 9012- Mokshata Chavan hereby declare that the work presented in this Field Project report Data Visualization Of Public Health And Environmental Data is an authentic record of our own work carried out under supervision of Prof. Tejal Wagh.

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CERTIFICATE

This is to certify that the above statement made by the students are correct to the best of my knowledge and belief.

Name and Signature of the Supervisor

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1. Introduction

Our field project on Data Visualization of Public Health and Environmental Data was carried out with a focus on Natural diseases. The main purpose of this study was to understand how diseases influenced by environmental factors can be studied, represented, and interpreted through data visualization techniques. The objective was not only to analyze the connection between environment and health but also to present findings in a way that makes the information clear and useful for decision-making in public health.

The main idea behind choosing this topic was to understand how health issues connected with nature and the environment can be studied, analyzed, and represented in a meaningful way. Through this work, we tried to explore how visual tools and methods can make complex health and environmental data easier to understand for researchers, doctors, and even the general public.

We also realized that data visualization is not only helpful for doctors and health workers but also for creating awareness among people. It can show patterns, like how some diseases increase during particular seasons, or how pollution and stagnant water contribute to health problems. This experience gave us practical knowledge of how important it is to collect accurate data and present it in a way that is useful for both planning and prevention.

Purpose of the Visit:

- To study how natural diseases are influenced by environmental factors like water, sanitation, and pollution.
- To learn how data about these diseases can be collected and shown using charts, tables, and graphs.
- To understand how visualization makes data easier to read and useful for doctors, health workers, and the community.

Why Natural Diseases?

- Common diseases like malaria, dengue, cholera, and respiratory illnesses are closely connected with environmental conditions.
- These diseases are major health problems in many parts of our country, especially during seasonal changes.
- Studying them helps us see the real link between environment and public health.

Learning Outcomes

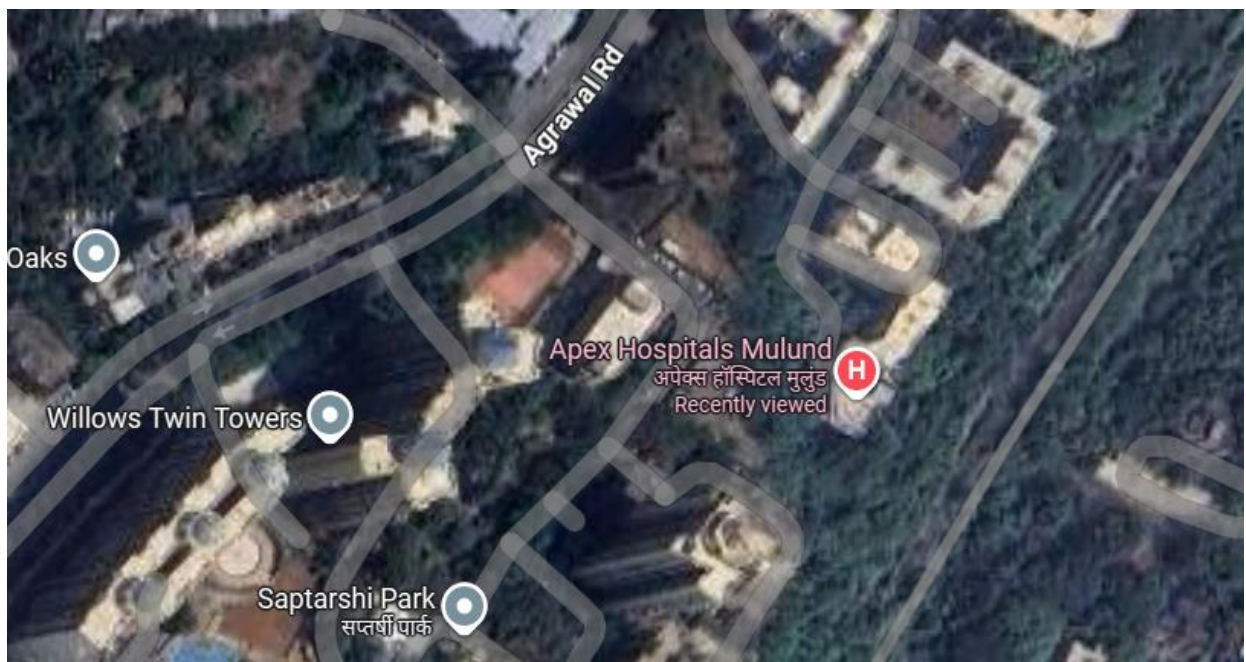
- We got practical knowledge of how data is collected and presented in a hospital setting.
- We understood the importance of visualization for finding patterns, such as seasonal increase in some diseases.
- The visit improved our observation, teamwork, and data-handling skills.
- It showed us how theoretical knowledge can be applied in practical situations

2. Date and Location

Date: The fieldwork was conducted on 21st August 2025 to 17th September 2025

Location: The visit was carried out at Apex Hospital, Mulund (West), Mumbai. The hospital is located in an urban area with a high population density, where environmental factors such as sanitation, water storage, and pollution directly influence public health.

This location gave us the chance to observe how hospital handle data related to natural diseases and how the surrounding environment contributes to the spread of illnesses like malaria, dengue, and respiratory infections. The setting helped us connect the role of environment with the patterns of diseases observed in the community.



3. Methodology

Approach and Tools: For our field visit, we mainly used observation and interaction as our approach. We visited Apex Hospital, Mulund (West), where we observed how data related to natural diseases is recorded and maintained. We also looked at how the environment around us, like sanitation, water storage, and pollution, plays a role in the spread of diseases. Some information was noted from hospital records and by talking to staff. Later, we arranged this data properly and made graphs and charts to understand the patterns more clearly. Simple tools like bar graphs and pie charts helped us to show the information in an easy way.

- To collect information, we spoke to the hospital staff and noted important points.
- We also referred to hospital records and available data about common diseases like malaria, dengue, and respiratory problem.
- we collected the information, we organized it carefully and converted it into visual forms like bar graphs, pie charts, and power BI tables.

The subsequent analytical phase employed descriptive statistical techniques to calculate frequencies, percentages, and average scores, enabling a comparative analysis to identify and quantify disparities in disease perceptions and priorities between the professional and public cohorts.

Finally, the visualization phase leveraged tools like Excel and Power BI to translate these analytical findings into a suite of clear, intuitive charts—including divergent bar charts to highlight perception gaps and heat maps for geographical trends—ensuring the results are accessible for both public health planning and community awareness initiatives.

Team Work And Supervisor Guidance: -

- Our fieldwork was conducted with total 5 student group was under the guidance of our supervisor DR. SARITA BAMBLE.
- One of our member was assigned roles such as note-taking explanation of every every department notes were taken.
- while other two member were assigned for observation of every department survey of patient and many more things.
- Rest two were assigned for data collection, sorting data of 200 peoples and analysis and data visualize them imto charts, graphs, and power BI

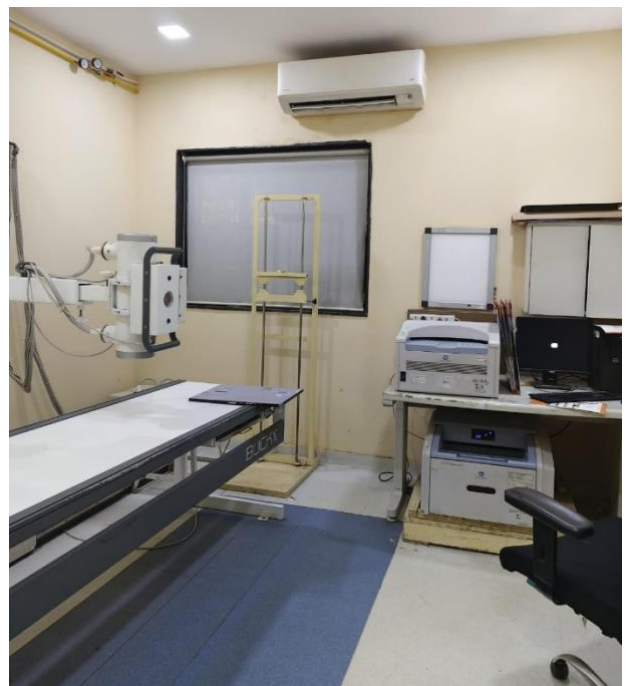
The supervisor provided guidance throughout the visit and helped us connect theoretical knowledge with practical understanding. Teamwork ensured that all aspects of the fieldwork—from data gathering to visualization—were covered efficiently

Our supervisor played an important role during the field visit. Right from the beginning, the supervisor explained the purpose of the fieldwork and guided us on what to observe at the hospital. They helped us in asking the right questions to the staff and also suggested how to take proper notes. The supervisor also explained how the collected information should later be arranged and visualized in graphs and charts. Whenever we had doubts, the supervisor cleared them and made sure we connected our practical observations with the concepts we had studied in class. Overall, their guidance gave us clarity and confidence in completing our field project.

4. Observations & Data Collection

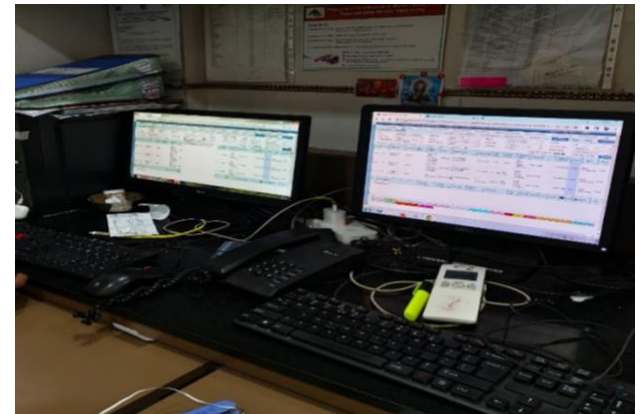
1. OBSERVATION FROM ECG/CT SCAN XRAY DEPARTMENT:

During our fieldwork at Mulund Apex Hospital, we collected data from the ECG, CT Scan, and X-Ray departments. In the ECG department, we observed common cardiac-related conditions and recorded patterns in heart abnormalities. The CT Scan department provided insights into internal organ conditions, injury cases, and disease prevalence, while the X-Ray department helped identify bone, chest, and respiratory-related issues. Data from these departments was organized and visualized using graphs and Power BI, allowing us to analyze trends and frequency of cases efficiently.



2. OBSERVATION FROM BLOOD LABRRATORY DEPARTMENT

The Blood Laboratory Department conducts a wide range of tests to assess overall health and diagnose various conditions. During the observation period, routine tests such as complete blood count, blood sugar, kidney and liver function tests, and lipid profiles were analyzed. The findings revealed that a significant number of patients had **anemia**, indicating low hemoglobin levels, while others showed signs of **infections** or inflammatory conditions. Elevated blood sugar and cholesterol levels were also noted in several cases, highlighting common metabolic disorders. The department plays a critical role in early detection of diseases, monitoring ongoing treatments, and providing essential diagnostic information that supports effective patient care.



3. OBSERVATION FROM NICU DEPARTMENT

- The NICU (Neonatal Intensive Care Unit) is dedicated to caring for critically ill or premature newborns. During observation, several cases were noted, including preterm infants, low birth weight babies, respiratory distress, jaundice, and infections. Continuous monitoring of vital signs, oxygen support, and timely medical interventions were essential in managing these cases.

- It was observed that early detection and prompt treatment significantly improved the chances of recovery. The department also emphasizes parental involvement and counseling, ensuring families are educated about newborn care and post-discharge follow-up. Overall, the NICU plays a vital role in reducing neonatal morbidity and



mortality through specialized care and constant supervision. The NICU employs advanced medical equipment such as incubators, ventilators, phototherapy units, and monitoring devices to support vital functions and maintain stable conditions for these infants. Nurses and doctors perform continuous monitoring of vital signs, regular feeding through IV or tube feeding, and timely administration of medications to ensure proper growth and recovery.

4. OBSERVATION FROM DIALYSIS DEPARTMENT:

A visit to the Dialysis Department reveals critical insights into the burden of chronic and end-stage kidney disease within the community. This department's operations serve as direct indicator of long-term public health challenges, often linked to preventable conditions.

Primary Service:

Provides life-sustaining treatment (hemodialysis or peritoneal dialysis) for patients with **severely impaired kidney function**, often due to:

Uncontrolled Diabetes: Diabetic nephropathy is a leading cause of kidney failure.

Unmanaged Hypertension: Chronic high blood pressure can -damage kidneys over time.

Other Causes: Glomerulonephritis, polycystic kidney disease, or chronic infections



5. OBSERVATION FROM INTENSIVE CARE UNIT (ICU) DEPARTMENT:

The In-Patient Department (IPD) is one of the most important sections of a hospital where patients are admitted for treatment, observation, or surgery and are required to stay for a specific period of time under continuous medical supervision.

Unlike the Out-Patient Department (OPD), where patients come only for consultation or minor procedures, the IPD is designed for those who need round-the-clock care and advanced facilities

The IPD consists of various wards and units such as general wards, private or semi-private rooms, maternity wards, pediatric wards, surgical wards, and intensive care units (ICU/NICU)

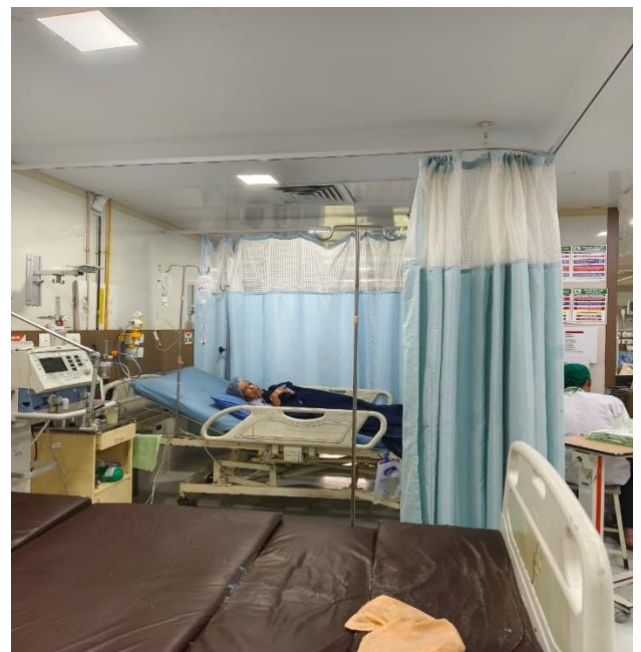


6. OBSERVATION FROM INTENSIVE CARE UNIT (ICU) DEPARTMENT:

The Intensive Care Unit (ICU) is a highly specialized section of a hospital designed to provide advanced medical care and continuous monitoring for critically ill patients. It is one of the most important departments in modern healthcare, as it deals with life-threatening conditions that require immediate attention and expert management.

Patients who are admitted to the ICU often suffer from severe illnesses, major accidents, organ failure, post-operative complications, or other medical emergencies that cannot be managed in a general ward.

The ICU is equipped with state-of-the-art facilities such as ventilators, cardiac monitors, defibrillators, infusion pumps, dialysis machines, and advanced diagnostic tools. These allow healthcare professionals to continuously observe and support vital functions like breathing, heart rate, blood pressure, and oxygen levels.

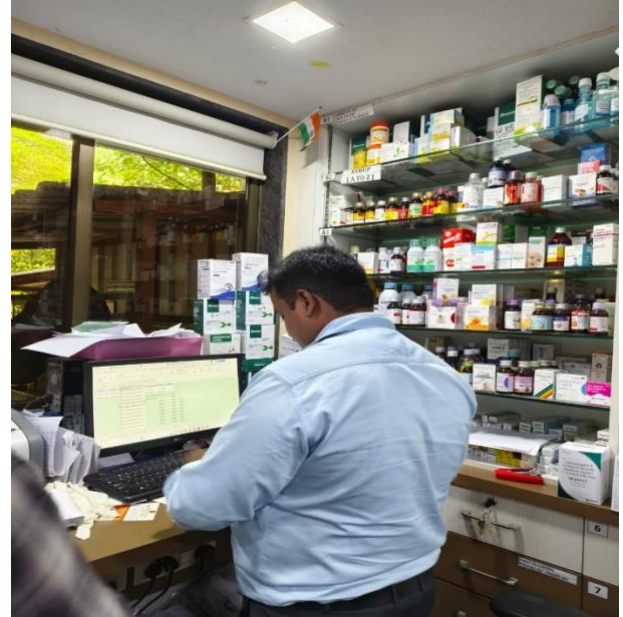


A QUICK SURVEY OF AN ICU PATIENT ASKING ABOUT THERE REQUIREMENT FROM HOSPITAL , MEDICLAIM EXPECTATION IF HAVE , THEIR NEEDS FROM HOSPITAL ABOUT TREATMENT ETC...

7. OBSERVATION FROM PHARMACY DEPARTMENT :

The Pharmacy Department is an important part of every hospital and healthcare system. Its main responsibility is to handle everything related to medicines – from storage and supply to safe usage by patients. The department makes sure that the right medicine is given to the right patient, in the correct dose, and at the right time. This helps in improving treatment outcomes and ensures patient safety.

One of the key functions of the pharmacy department is to store and manage medicines. It keeps a proper stock of essential, emergency, and life-saving drugs so that they are available

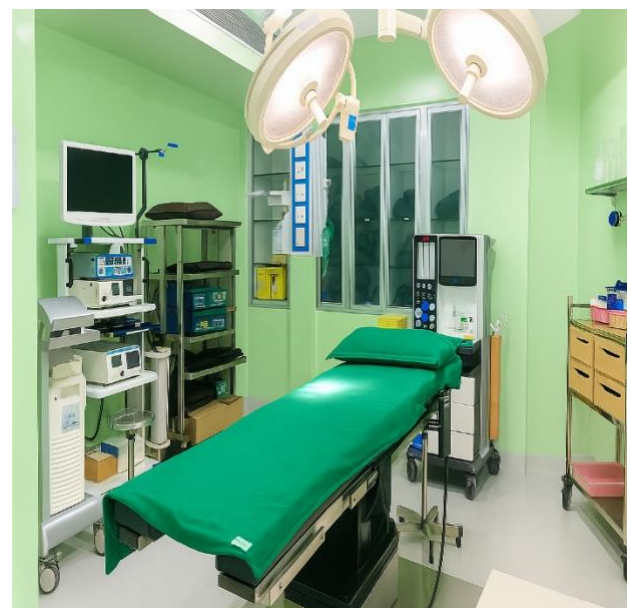


8. OPERATION THEATRE (OT) DEPARTMENT:

The Operation Theatre (OT) is a specialized facility in a hospital where surgical procedures are carried out under strict sterile and controlled conditions. It is one of the most important parts of a hospital, as it allows doctors and surgeons to perform operations ranging from minor procedures to complex and life-saving surgeries.

The OT is designed to maintain a sterile environment, preventing infection and ensuring patient safety during surgery.

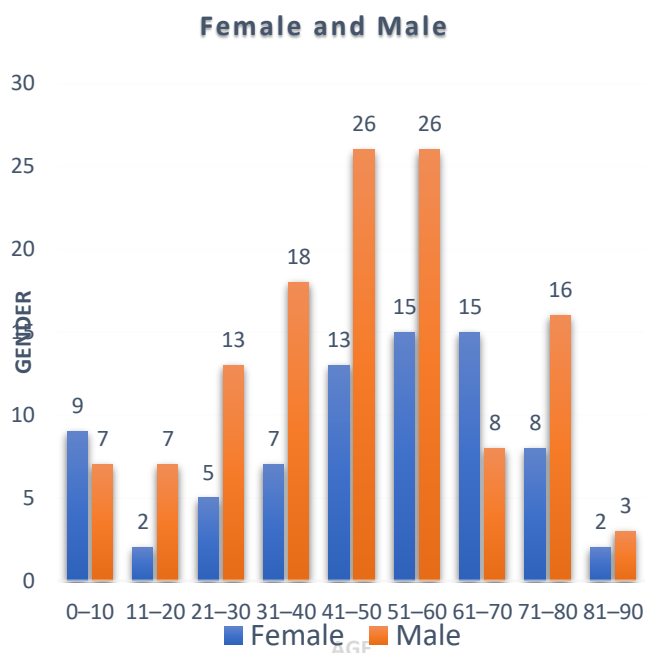
An operation theatre is equipped with advanced surgical instruments, anesthesia machines, operating tables, surgical lights, monitoring devices, and sterilization equipment. It is supported by a team of highly trained professionals including surgeons, anesthesiologists, nurses, and technicians who work together to ensure that the surgical process is carried out smoothly and safely



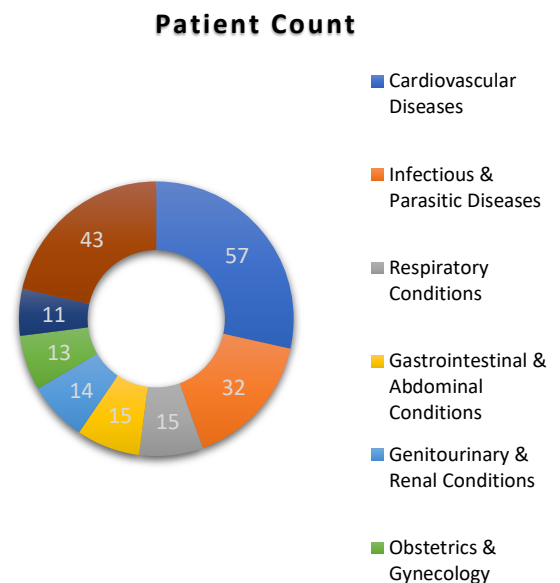
DATA COLLECTION:

The data collected during the field visit has been systematically analyzed to understand the distribution of patients across age groups and gender in the various department. we'll see some Graphical representations such as pie charts and bar charts have been used to provide a clear visual understanding of the patterns observed.

- All of this below bar graphs, pie charts, line charts etc. contains data of three months (July, August, September)

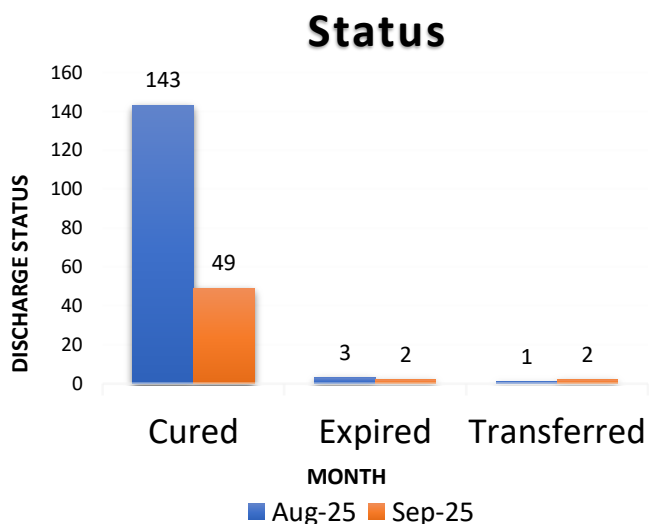


[No. of patient by age and gender]

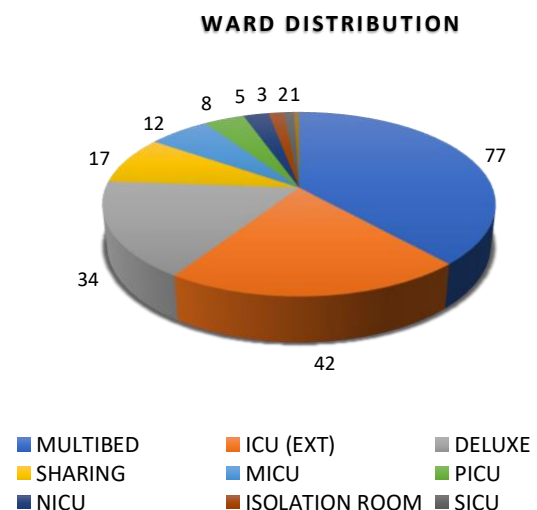


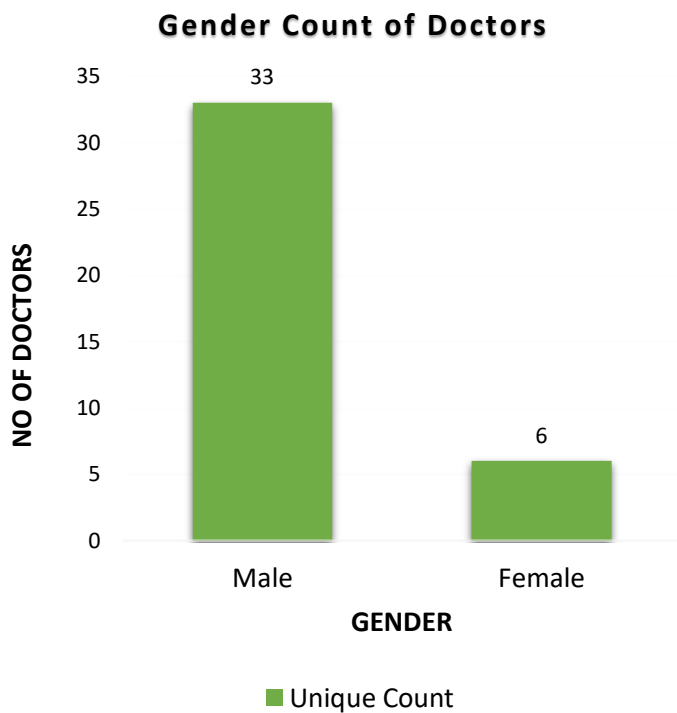
[No. of patient by age and gender]

No. of patient expired, Transferred]

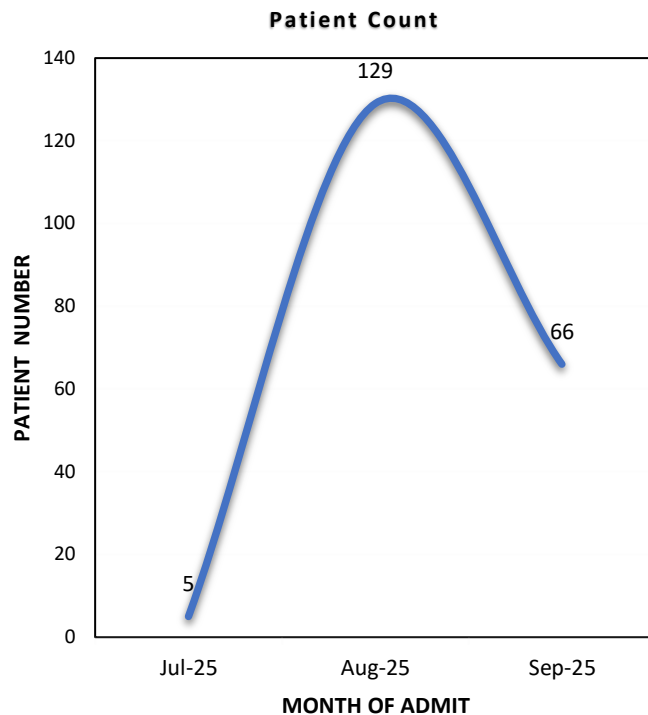


[No. of departments]





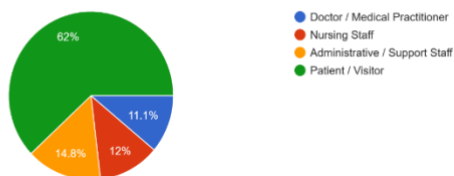
[No. of male and female doctors]



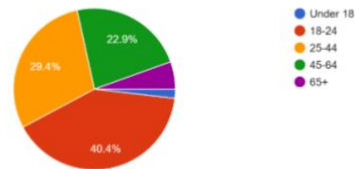
[No. of patients admitted in jul, aug, sept]

HEALTH CARE SURVEY

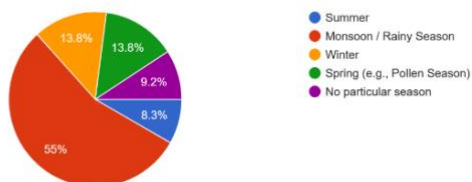
1. ROLE:
108 responses



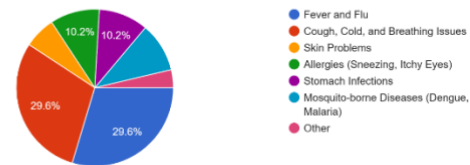
2. Age Group:
109 responses



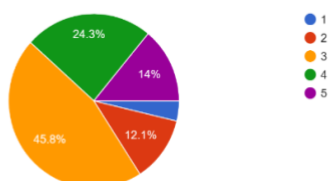
3. In your opinion, which season has the most significant impact on people's health here?
109 responses



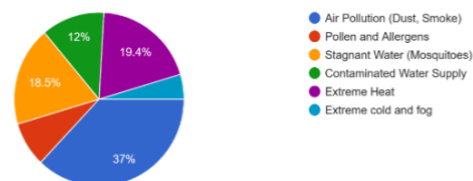
4. What is the most common type of illness you associate with that season?
108 responses



5. On a scale of 1 to 5, how much do you agree with this statement: "The air quality in our area affects my/patients' health."
107 responses

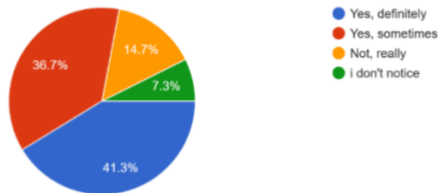


6. From your experience, what are the two biggest environmental factors causing health issues here?
108 responses



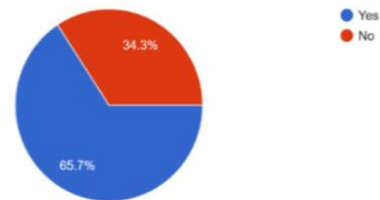
7. Have you noticed a clear increase in health problems linked to a specific environmental change (e.g., after rains, during crop burning, during a heatwave)?

109 responses



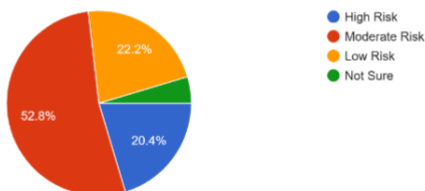
8. Do you take any specific preventive measures based on the season?

108 responses



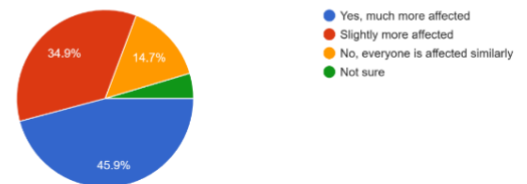
9. How would you rate the environment around your home/workplace in terms of potential health risks? (e.g., mosquitoes, dust, mold)

108 responses



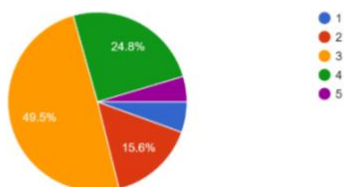
10. In your experience, are children and the elderly significantly more affected by these seasonal illnesses than young adults?

109 responses



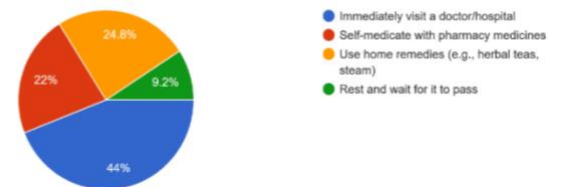
11. How would you rate the overall burden of seasonal diseases on you/your household/this facility?

109 responses



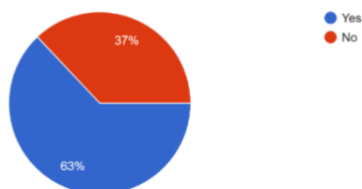
12. When you experience a significant seasonal illness, what is your first course of action?

109 responses



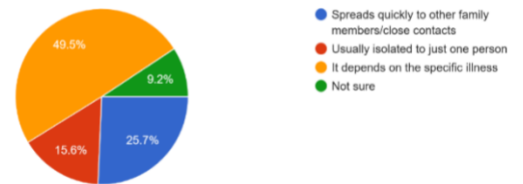
13. Does your area experience problems with stagnant water (e.g., uncovered pots, clogged drains, potholes) which can lead to mosquitoes?

108 responses

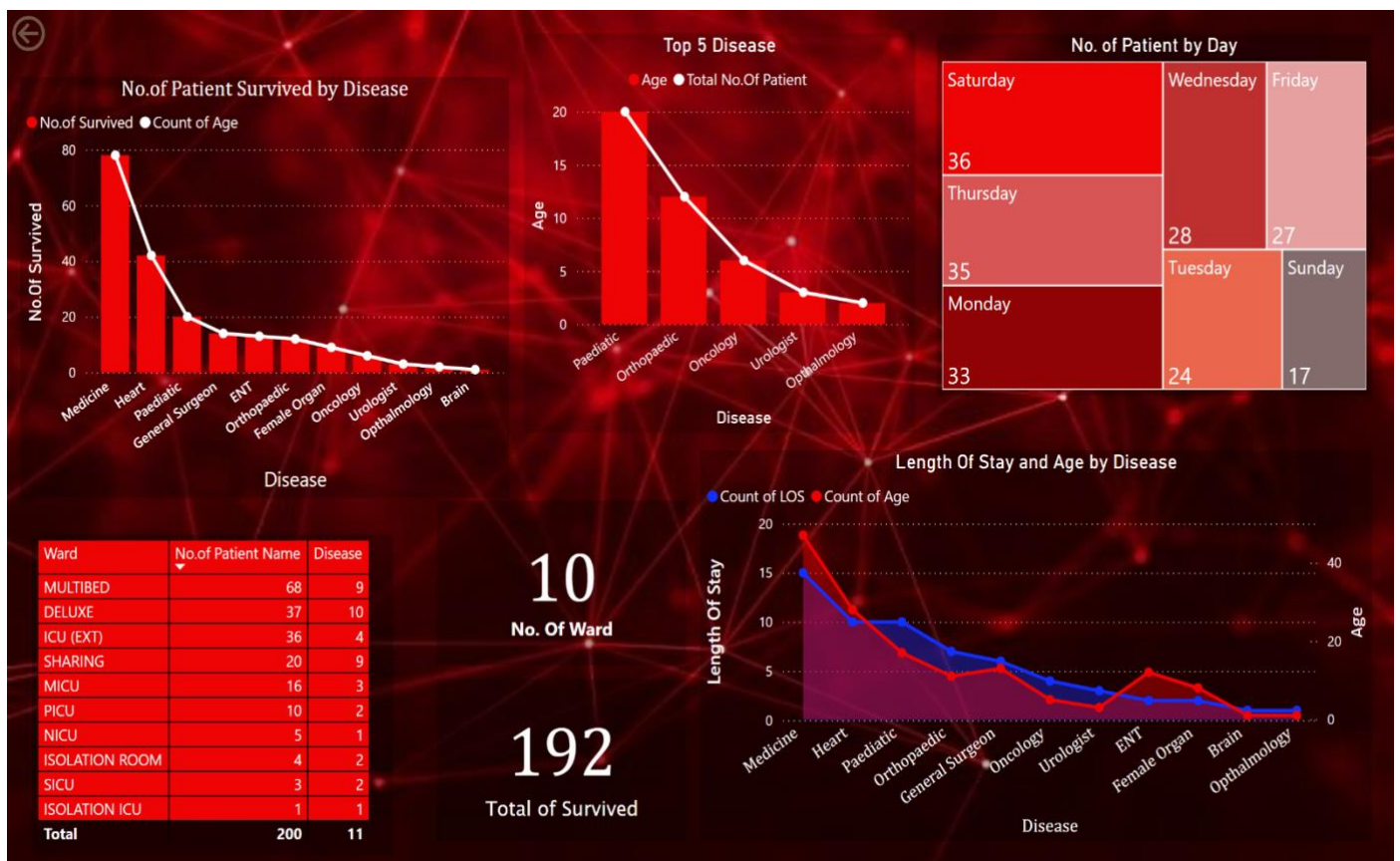
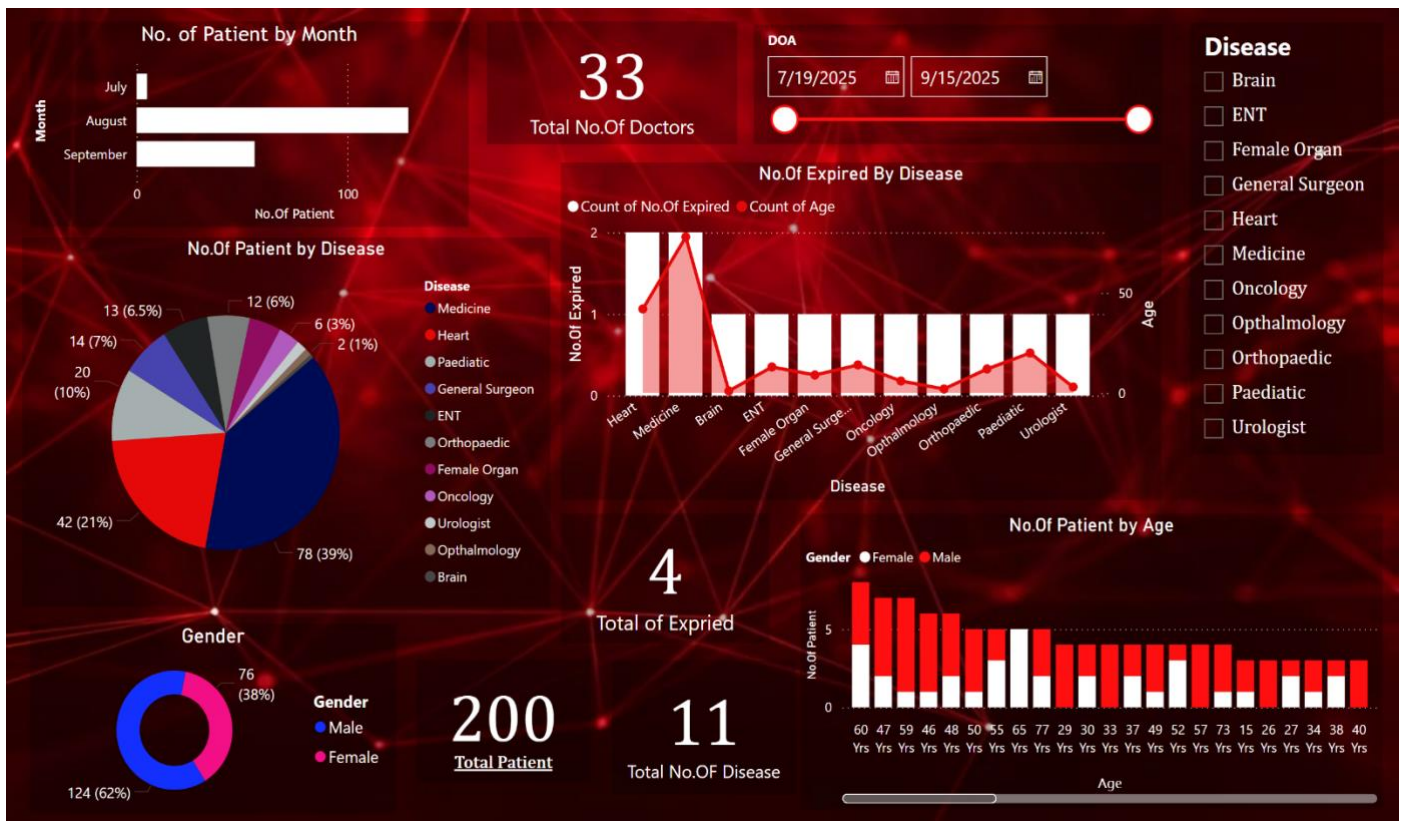


14. When a family member/patient gets sick with a seasonal illness, how does it typically spread within your household/the community?

109 responses



DATA ANALYZATION THROUGH POWER BI



5. Analysis & Findings

During our fieldwork, we collected and studied data from different departments, which gave us a better picture of the health issues commonly faced by patients. In the blood laboratory, the most frequent findings were low hemoglobin levels (anemia), high blood sugar, and high cholesterol. These results show that lifestyle-related problems like poor diet, lack of physical activity, and stress are major contributors to health conditions in the community. Such conditions, if not detected early, can lead to bigger health issues like heart disease or kidney failure.

Data visualization helped us clearly understand these trends by organizing the information in a way that showed patterns, frequency, and risk groups. This supports the objective of our fieldwork, as it allowed us to interpret complex hospital data in a simpler and more useful manner. From the data collected during our field visit at Apex Hospital, we observed clear links between environmental conditions and the occurrence of natural diseases. By organizing the information into graphs and charts, we were able to identify patterns and trends that showed how different factors influenced the spread of illnesses.

Key Findings:

- Diseases like malaria and dengue showed a clear rise during the monsoon season due to waterlogging and mosquito breeding.
- Respiratory issues like cases of asthma and other breathing problems were more common in areas with high air pollution and during seasonal climate changes.
- Water-Borne Disease illnesses like cholera and diarrheal infections were found to be connected to sanitation issues and the use of contaminated water.
- Representing the data through bar graphs and pie charts helped us to compare the occurrence of different diseases and understand their relationship with environmental conditions.
- We noticed that hospitals collect a lot of raw data, but visualization makes it more meaningful, helping health workers and authorities plan preventive measures.
- The findings proved that environmental factors have a major role in the spread of natural diseases, and proper visualization is a useful tool to highlight this connection.

Our fieldwork helped us understand the real situation of patients by directly studying different departments. The blood laboratory showed that anemia, high sugar, and cholesterol were the most common findings. These are conditions that usually develop slowly but create a huge impact on overall health if ignored. They also point toward lifestyle patterns of people, such as unhealthy eating habits, lack of exercise, and stress. This connects with the growing burden of non-communicable diseases, which are now one of the biggest public health concerns.

Challenges Faced:

While doing the fieldwork, we came across several challenges:-

- Collecting data from multiple departments was time-consuming because doctors and staff were busy with their duties, and we had to wait for their availability.
- At times, the records we got were incomplete or delayed, which slowed down our analysis. A
- Another challenge was handling such a large amount of data without making mistakes, since even a small error could affect the final results.
- We also had to be very careful about maintaining patient confidentiality.

In some cases, we were not given full access to all the information we needed, and this became a limitation in our analysis. Technical difficulties, such as arranging records from different departments in a proper format, also made the process more complicated.

Despite these challenges, we managed to carry out the fieldwork successfully with the guidance of our supervisor. Their support helped us stay systematic, double-check our data, and overcome the difficulties we faced. These challenges taught us the importance of patience, teamwork, and proper planning in research work.

These challenges taught us that fieldwork is not just about data collection, but also about adjusting to real-life situations, solving problems, and staying patient. It also made us realize that accurate data, ethical handling, and teamwork are the backbone of any successful research.

6. Conclusions

Through this fieldwork, we were able to understand how different departments in a hospital contribute to the overall care of patients and how data visualization can make this understanding clearer.

The information we collected showed that lifestyle-related diseases like diabetes, hypertension, heart problems, and kidney issues are becoming more common, and at the same time, sensitive areas like newborn care require advanced facilities and constant attention. By converting the data into graphs and visuals, we could see patterns more easily, which helped in identifying the areas where more awareness, prevention, and treatment efforts are required.

The fieldwork experience gave us a very clear understanding of how different hospital departments work together to provide complete care to patients, and how data can be used in a powerful way when properly collected and visualized. By visiting departments like the laboratory, dialysis unit, NICU, and diagnostic sections, we learned that every area has its own importance and contributes differently to patient health.

The blood laboratory highlighted the rising cases of lifestyle-related diseases such as high sugar, cholesterol, and anemia, while the dialysis unit showed us the long-term impact of uncontrolled diabetes and hypertension. Similarly, the ECG, X-ray, and CT scan departments revealed the importance of modern diagnostic tools in detecting health problems early, and the NICU gave us an insight into how delicate neonatal care is and how advanced facilities can save newborn lives.

When we analyzed all this data, it became clear that non-communicable diseases and neonatal health are two major concerns today. At the same time, we realized that raw numbers and records alone cannot explain the full story. Once the data was visualized into graphs and charts, the results became much clearer and easier to understand. Patterns such as which diseases were most common, which age groups were affected, and what complications were frequent became visible immediately. This made us appreciate the true value of data visualization as a tool in public health research.

This experience also taught us the value of teamwork, discipline, and guidance. It was not only about observing departments but also about learning the importance of accuracy, confidentiality, and proper communication. Even though we faced challenges during data collection, we learned to overcome them through planning and cooperation.

Overall, this fieldwork made us realize that public health and hospital care are deeply connected with data, and when data is properly managed and visualized, it becomes a powerful tool to improve healthcare services and decision-making. It was a very practical and meaningful learning experience that gave us confidence and a better understanding of real-world health problems.

7. Recommendations

- From our experience, we realized that fieldwork can be made more effective with better planning and use of proper tools. One important recommendation is to prepare a clear data collection format before visiting departments, so that information can be recorded in a uniform way without missing details.
- More time should also be allocated to each department, as sometimes we felt rushed due to the busy hospital environment. The use of advanced tools like Power BI and other data analysis software should be expanded because they make the visualization part faster and more accurate.
- Team members should also be given proper training in handling large amounts of data and ensuring accuracy while recording. Another important recommendation is to create stronger communication with hospital staff before starting fieldwork.
- This can save time and reduce delays while accessing records. Lastly, dividing roles clearly among team members can help avoid confusion and improve efficiency during the fieldwork process.
- Based on our findings, the next step should be to expand the research beyond one hospital and compare data from multiple healthcare centers. This would help to confirm whether the trends we observed, such as the high burden of non-communicable diseases and the sensitive nature of neonatal care, are similar across different regions.
- More detailed studies should also be done on preventive measures for lifestyle-related diseases, as our data showed how badly they affect patients in the long term. Another next step could be to use data visualization not only for analysis but also for awareness programs, where patients and the general public can understand health issues better through simple charts and visuals.
- Regular follow-up studies should also be carried out to see if the health patterns are changing over time. By continuing this research and expanding it further, we can contribute more strongly to the improvement of public health and hospital services.
- The use of advanced digital tools like Power BI, Tableau, or even AI-based analysis can be included to make the visualization more professional and impactful.

8. Appendices

- Along with our report, we have attached some extra materials that make our work clearer and more reliable.
- These include tables, graphs, and charts made from the data we collected during the fieldwork. They help to easily see the patterns and understand the results.
- We have also kept the raw data sheets so that anyone can check how the information was recorded. Wherever possible, we have added photographs of departments and field visits to give a real view of our experience.
- In some cases, maps or layouts can also be included to show the working setup of departments.

By adding these documents, our report becomes more complete. It not only explains the findings but also gives proof and visuals that support our observations. This makes the work more trustworthy and easy to follow.

END OF REPORT